Real-user queries

500
Open-ended questions q

① Frozen evaluation inputs

Question-specific rubric
Positive budgets: 20 / 35 / 30 / 15
Negative triggers, groups, caps

System configuration
Model + harness
+ reasoning effort

Frozen outputs
Response a
Execution Trace τ

② Two-stage evidence audit

Stage 1: Provenance

Response a

Claim extraction
Atomic claims

Trace τ

Supported by Trace?

yes → **Grounded**

no → Stage 2

Four outcomes retained for diagnosis

Stage 2: Independent verification

Independent external evidence

- supported → **Verified but ungrounded**
- undetermined → **Unverified**
- refuted → **Hallucinated**

Claim + refuting evidence

③ Quality scoring

Inputs: response a + frozen rubric

Positive and other negative items
Binary judgments from the response

Factual-reliability negative items
Apply frozen trigger conditions to audit evidence

Positive score − effective penalties
Max per root cause · credit positive losses · apply caps

Quality score Q